

SOCIAL IDENTITY AND LEARNING: BELIEFS AND PREFERENCES

Akshay Moorthy*

September 20, 2026

Abstract

This paper examines how the identity of an information source shapes learning in settings where decision-makers and information sources share the same incentives and information. The analysis draws on three large-scale online experiments designed to disentangle identity preferences from beliefs about the quality of an information source. The experiments are conducted with both naturally occurring identities (caste and religion in India) and experimentally induced identities. Across identity contexts, there is little evidence that preferences for the identity of an information source influence social learning when information quality is known. On the other hand, beliefs about information quality strongly influence learning, but participants are overconfident and often do not learn when it would benefit them. The results show that source identity plays a limited role in information adoption when information about source quality is available, and highlight the importance of providing credible signals of information quality when social identities are salient.

Keywords: Social learning, social identity, discrimination

JEL Codes: D83, D91, J71

*Moorthy: Faculty of Business and Economics, University of Lausanne, Switzerland; akshayarun.moorthy@unil.ch; ORCID: 0000-0001-8422-3145.

1 Introduction

Social learning is fundamental to human evolutionary success (Boyd et al., 2011; Henrich, 2016) and influences many economic decisions (Mobius and Rosenblat, 2014; Bikhchandani et al., 2024). In social learning, decision-makers acquire information by observing others’ actions and decide how to use that information. When decision-makers observe another person’s actions, they may also notice components of that person’s social identity such as their race, religion, or gender, which may influence how they learn. The links between identity and learning have received attention, with research showing that people are more likely to respond to information from sources perceived as socially closer or ideologically aligned (Dolan et al., 2012; Maclean et al., 2019; Robbett et al., 2023; García-Hombrados et al., 2026). Recent work has begun to investigate the mechanisms behind these effects (Zhang and Rand, 2023; Dekel and Shayo, 2025; Gangadharan et al., 2025; Bauer et al., 2026), but it remains unclear whether such differences reflect beliefs about the quality of information provided by sources of different identities or preferences over the identities of those information sources.

This paper examines how these two channels shape learning. The distinction between these channels – which parallels statistical and taste-based discrimination (Phelps, 1972; Arrow, 1973; Becker, 1957; Guryan and Charles, 2013) – matters for information delivery and policy design. When identity is perceived to be informative about signal quality, conditioning on identity could improve learning and decision-making, while misspecified beliefs could impair decision-making. By contrast, discounting or overweighting information because of identity-based preferences could impair accurate belief formation. This preference channel could lead to information frictions within firms and teams (Itoh and Morita, 2023), ineffective policy communication (Wabitsch, 2024), or belief polarisation (Levy and Razin, 2019; Bikhchandani et al., 2024; Robbett et al., 2023; Bauer et al., 2026). Therefore, understanding the roles of identity-driven beliefs and preferences in social learning is crucial for designing communication strategies and information interventions targeted at improving decision-making.

This paper reports results from three incentivised online experiments (with a total of 1,982 participants) that are designed to disentangle beliefs about information quality from preferences for the information source’s identity. Two experiments study naturally occurring identity contexts, caste and religion in India, and the third experiment studies experimentally induced identity. The experiments share the same structure: a two-period task in which participants make incentivised decisions in each period. In the first period, participants make an independent decision in a cognitively demanding “balls-and-urns” task – a well-established paradigm for examining belief formation and learning (Benjamin, 2019; Weizsäcker, 2010) – in which calculating the correct posterior probability is required to earn an incentive. In the second period, participants are shown a probability estimate made by another individual – the *source* – on the same task. Participants then decide whether to learn from the source by *switching* to the source’s estimate, or not to learn by *sticking* to their independent estimate. This is the main outcome variable. Together, the three experiments provide a dataset of $\approx 11,900$ switching decisions.

The empirical strategy relies on controlling beliefs about the accuracy of the source’s decision while varying the source’s identity. This is achieved by experimentally assigning two features of the decision-making context in the second period. First, participants receive a randomly assigned signal

of the *quality* of the source’s decision. This is the probability that the shown decision is correct or, equivalently, the probability with which switching to the source would yield the incentive. Second, they observe the source’s social *identity*, which is also randomly assigned and varies between treatment conditions in each experiment. Under the identifying assumption that the perceived quality of the source’s decision is independent of the source’s identity, comparing switching behaviour across identity treatments reveals preferences for learning from sources belonging to one identity relative to the other when information quality is experimentally provided. The controlled setting allows the isolation of the effect of source identity on learning by holding source quality fixed and ruling out other channels through which identity may affect decisions, such as signalling, reputation, and social image concerns. Source identity is also present in settings that involve these channels, although the design does not speak to how its effect interacts with these other dimensions to influence learning.

This experimental framework also sheds light on a form of learning in which decision-makers choose to learn from others despite having access to identical information. Both the decision-maker and the source make choices on the same task, operate under identical incentive structures, and possess the same information, which is sufficient to determine the incentive-maximising decision. Consequently, a decision to switch in the second period cannot be attributed to a belief that the source possesses superior or different information, which is usually the focus of social learning experiments. In this case, switching may reflect a belief that the source has a better interpretation of the same information. This channel of learning could be relevant in many real-world decisions, which often involve seeking not only better information but also more effective interpretations. For instance, patients seek a second opinion from a doctor who observes the same test results, or individuals follow peers’ choices of insurance or investment plans despite having access to the same product and market information.

Two experiments study naturally occurring identities, caste and religion in India. These factors play a central role in Indian social and economic life (Munshi, 2019; Mosse, 2020; Iyer, 2016), and there are substantial differences in income and educational attainment between caste and religious groups (Census of India, 2011; Asher et al., 2024). A person’s religion or caste is often salient as it can be reliably inferred from their name, mode of speech, body language, or other visible markers. Prior research has provided evidence that caste and religion influence decision-making through both statistical and taste-based channels. Together, these factors suggest that caste and religious identities could influence learning through both preferences and beliefs.

In experiment *Caste*, the source belongs to either a high-status or low-status caste group. In experiment *Religion*, the source is either Hindu or Muslim. These experiments were conducted online and participants were recruited through panel survey providers. Caste and religious identity are signalled by assigning surnames to the source, following a strategy that has been used in many different contexts (Guryan and Charles, 2013; Bertrand and Duflo, 2017) including caste (Hoff et al., 2011; Sankaran et al., 2017) and religion (Fershtman and Gneezy, 2001; Chakravarty et al., 2016). As participants in the sample have both religious and caste identities, the design allows for the analysis of general preferences (whether the decision-maker learns more from a source of a particular caste or religion), in-group preferences (whether the decision-maker learns differently from a source sharing their religious or caste identity), and heterogeneity across overlapping identities (whether

preferences vary by different combinations of caste and religious identities).

Experiment *Minimal* complements the natural-identity experiments by providing a controlled examination of the link between identity and learning without invoking many of the empirical challenges associated with naturally occurring identities (Charness and Chen, 2020). The experiment is based on the minimal group paradigm (Tajfel and Turner, 1979; Chen and Li, 2009), in which identity is experimentally induced – participants are assigned to one of two groups based on choices in an unrelated task, and see a source from either their in- or out-group. Participants in the *Minimal* experiment were recruited via an online survey platform with participants from the EU, UK, and the US. These three experiments are complemented by two supporting experiments and a separate survey focused on eliciting incentivised beliefs about the performance of different caste and religious groups on the experimental task.

The first main finding is that participants are responsive to the randomly assigned quality signal: the propensity to switch increases with information quality. Across experiments, participants choose to switch to the source in 24 – 32% of decisions, showing that participants choose to learn even though sources have the same information. However, most participants leave money on the table by not switching even when the source is high-quality. This behaviour is consistent with overconfidence – more confident participants switch less, and overall, individuals overestimate the accuracy of their initial decisions. These patterns are qualitatively consistent across all samples and identity contexts, though responsiveness to information quality varies. A 40 percentage point (pp) increase in the probability that the source is correct (the range of experimental variation) leads to a ≈ 5 pp increase in switching in the *Caste* and *Religion* experiments, and a ≈ 25 pp increase in the *Minimal* experiment.

At the same time, there are substantial differences in underlying beliefs about the performance of different identity groups: Hindus believe that high-status caste Hindus outperform low-status caste Hindus by about 7 pp, and Muslims by about 15 pp. A supporting experiment (in which the quality signal was not provided) provides suggestive evidence that such beliefs serve as proxies for information quality.

The second main finding is that there is little evidence that participants have a preference for learning from a source with a particular identity in the *Caste*, *Religion*, and *Minimal* experiments. There are no statistically significant differences in whether participants switch to a source from a high- or low-status caste group in experiment *Caste*, a Hindu or a Muslim source in experiment *Religion*, or an experimentally induced in-group or out-group source in experiment *Minimal*. A battery of diagnostic analyses supports these null results, allaying concerns that the results are driven by inattention or a lack of comprehension. Statistical equivalence tests show that the results are informative for medium-sized effects, but do not rule out the possibility of small effects. A supporting experiment confirms that the experimental paradigm is equipped to detect differences in switching because of the representation or framing of the source, showing that participants switch more to sources labelled as a *Computer* than when the source is an anonymous human.

Additional analyses examining heterogeneity by task-specific moderators, exposure to religious and caste diversity, attitudes toward caste-based affirmative action, and individualism do not uncover a consistent subgroup or sample restriction in which source identity meaningfully affects switching behaviour. Analyses within the *Caste* and *Religion* experiments using participants' self-reported re-

ligious and caste identities do not reveal differences in switching behaviour when decision-makers and sources belong to the same identity group. Exploratory subsample analyses of overlaps between religious and caste identities in the *Religion* experiment show that SC/ST/OBC Hindus switch less to Hindu sources than to Muslim sources, while the treatment difference is significantly larger for General-caste Hindus than for SC/ST/OBC Hindus. There is no evidence of such heterogeneity among Muslim participants.

Taken together, the evidence suggests that participants do not have a systematic preference for the identity of an information source when source quality is known. The exploratory results for overlapping caste and religious identities suggest that preferences may emerge for particular combinations of participant and source identities, but these patterns are not consistent across samples or analyses.

This paper is related to the empirical social learning literature ([Mobius and Rosenblat, 2014](#); [Bikhchandani et al., 2024](#)) and contributes directly to a strand of this literature studying how the identity of an information source shapes information acquisition and processing. Prior experimental work has documented identity-based differences in information processing in a variety of settings: caste or social status in India ([Gangadharan et al., 2025](#)), gender differences in intra-household social learning ([Conlon et al., 2021](#)), political in-group biases ([Robbett et al., 2023](#); [Zhang and Rand, 2023](#)), national in-group following ([Dekel and Shayo, 2025](#)), minimal identity ([Berger et al., 2018](#); [Parys and Ash, 2018](#); [Zou and Xu, 2023](#)), endogenous source selection ([Dimant et al., 2024](#)), and agricultural technology adoption ([BenYishay and Mobarak, 2019](#)). Some of this research has explored the “messenger effect” ([Dolan et al., 2012](#); [Maclean et al., 2019](#); [Wabitsch, 2024](#); [García-Hombrados et al., 2026](#); [Afrouzi et al., 2024](#)). Together, these studies establish that source identity can influence social learning across a range of settings. The main contribution of this paper is to distinguish the informational role of identity (the beliefs channel) from a conditional response to identity (the preference channel).

Three recent papers examine the mechanisms behind these effects more directly, and are closely related to the experiment developed in this paper. [Zhang and Rand \(2023\)](#) find partisan differences in responses to advice in a non-political task and use variation in accuracy incentives, elicited beliefs about partisan competence, and feedback to distinguish belief-based from preference-based explanations. [Bauer et al. \(2026\)](#) study the effects of partisan identity across different stages of belief formation, including information demand and belief updating, and identify a source-utility channel that remains relevant when processing information in a neutral task. [Dekel and Shayo \(2025\)](#) distinguish social learning from in-group conformity and out-group differentiation by experimentally varying observed in-group and out-group behaviour, and find stronger in-group following as well as out-group counteraction. This paper complements these studies by independently randomising source identity and information about decision-specific source quality. Despite substantial identity-linked beliefs about performance, I find little evidence that source identity affects learning once source quality is known.

This paper also speaks to the literature on social identity in India by examining how caste and religion shape learning. Caste and religion are important aspects of social life and economic behaviour in India (see [Iyer \(2016\)](#); [Munshi \(2019\)](#); [Mosse \(2020\)](#) for reviews), and a large body of research

has documented the influence of religious and caste identity in various decision-making domains such as consumption (Atkin et al., 2021), hiring in labour markets (Siddique, 2011), labour supply (Cassan et al., 2021; Oh, 2023), marriage markets (Banerjee et al., 2013), and teamwork (Ghosh, 2025). Closely related to this paper is the recent work by Gangadharan et al. (2025), who show that Indian farmers are more likely to follow advice from their caste in-group or high-status individuals. Relative to Gangadharan et al. (2025), the results in this paper suggest that preferences for the religious or caste identity of information sources do not appear to strongly affect learning when the quality of information is experimentally provided. The results also show systematic differences in beliefs about the abilities of different caste and religious groups, and suggest that these beliefs may be used to infer information quality when the latter is not observable.

Finally, the paper contributes to work on learning when observed actions contain no additional raw information. Agranov et al. (2026) show that observing others’ actions can improve decisions even when all underlying signals are public. Relatedly, Çelen et al. (2010) show that people respond differently to advice and observed actions even when the two are informationally equivalent. The experiments in this paper add source identity and an explicit measure of decision quality to this setting. The results show that while participants do benefit from others’ interpretations of common information, they fail to use high-quality information, a pattern consistent with overconfidence in their own initial choices.

The paper is organised as follows: Section 2 presents the experimental design, provides intuition for the identification and hypotheses, and describes the various experiments and identity contexts. Section 3 examines descriptive and experimental evidence on the role of beliefs. Section 4 presents an analysis of the preference channel. Section 5 concludes.

2 Experiment Design

The study consists of three main and two supporting experiments. Within an experiment, participants were randomly assigned upon entry to one of two treatment conditions which differed only in the identity of the information source. All experiments used the same task structure, but differed in the identities used and the manner in which the identities were cued.

This section first introduces the common experimental structure, presents a conceptual discussion, and finally describes the different identity contexts of each experiment along with relevant background information. The experiment instructions are presented in Appendix D.

2.1 Experimental task

The experimental task builds on the “balls-and-urns” paradigm, which is widely used to study belief formation and social learning. Participants complete six tasks, and each task takes place in two periods. Participants make an incentivised decision in each period.

First period. Participants are shown two “urns”, A and B, each of which contains 100 red or black coloured balls. A contains θ red balls, and B contains $100 - \theta$ red balls. One of the urns is randomly

selected and the participant does not know which urn is selected. Urn A is selected with probability p and urn B is selected with probability $(1 - p)$. k balls are drawn (with replacement) from the chosen urn and shown to the participant. Participants use a slider to make an incentivised estimate $y_1 \in [0, 100]$ of the probability that urn A was chosen. After making the decision, participants state their confidence that y_1 is within ± 2 percentage points of the correct decision. The tasks are predefined in the sense that the values of the base rate p , urn composition θ , and the signals (colours of k drawn balls) are the same for all participants. This design choice ensures that participants are exposed to tasks of the same complexity. Participants see tasks in a randomised order. The tasks, corresponding parameters, and other details are listed in Appendix Table B.1.

Participants can calculate the correct answer for each task by applying Bayes' rule using the provided information. Participants receive detailed instructions and work through a training task on how to use Bayes' rule to calculate the correct answer.¹ However, the task is also cognitively demanding, which means that learning from others may be beneficial. Participants have an opportunity to do so immediately after they make their first-period decision.

Second period and main outcome. Participants are shown a decision $y_s \in (0, 100)$ made by another person, the source s , on the same task. Participants know that the source had the same information and saw the same draws. Participants see (i) the group *identity* g of the source, which is randomly assigned between-subjects, and (ii) a signal Q of the *quality* of the decision. Figure 1 shows the representation of these elements in the experiment interface. Next, participants make an incentivised binary choice y_2 : whether to *stick* to their first decision ($y_2 = y_1$) or *switch* to the shown decision ($y_2 = y_s$). This is the main outcome variable, *Switch*, which is 1 when $y_2 = y_s$ and 0 when $y_2 = y_1$.

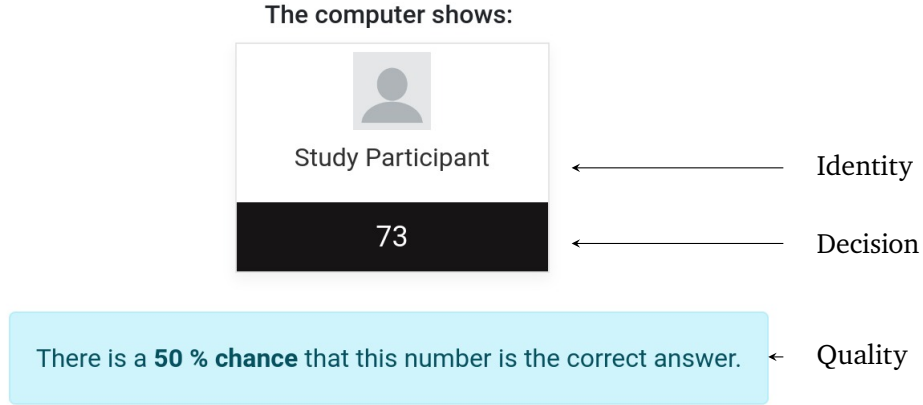
Eliciting learning behaviour using this discrete outcome rather than a continuous second-period estimate has two advantages. First, the binary decision clearly distinguishes between updates that are driven by a choice to move away from one's initial decision and updates driven by hedging or mechanical drift due to task complexity. Second, the discrete decision space is analogous to many real-world situations such as choosing a health insurance plan, and is simple to understand – a key consideration when deploying this experiment in multiple contexts. On the other hand, the design excludes the intensive margin of learning as participants cannot partially update their beliefs.²

Source decisions. Behind the scenes, each task is associated with two source decisions. One of these decisions is correct and the other is incorrect, and the same set of decisions is used within an experiment. These decisions were drawn from the incentivised first-period decisions made in separate experiments with participants from the same populations as the main experiments. At least one person from each identity group made each chosen decision. Appendix Table B.1 provides a list of the correct and incorrect source decisions for each task. Additional details on how the source

¹The correct answer can be calculated as follows: Let r denote the number of drawn red balls. Then, $y_T = 100 \times \frac{p\theta^r(100-\theta)^{k-r}}{p\theta^r(100-\theta)^{k-r} + (1-p)(100-\theta)^r\theta^{k-r}}$. At the time of data collection, LLM tools did not reliably solve this task.

²A supporting study conducted concurrently with the main experiment provides data to quantify this trade-off. This is discussed in Section 2.6.

Figure 1: Source – Attributes



Notes. The figure shows how the source is represented in the experiments. *Identity* is randomly assigned between-subjects within each experiment group. The *quality* of the source's decision is randomly assigned at the task level, and is the probability that the shown *decision* y_s is correct (earns the incentive).

decisions were selected are provided in the discussion of the separate experiments in Section 2.3 below and in Appendix C.1.

Quality signal. Participants are given a signal of the quality Q of the source's decision, which is the probability that y_s is the correct decision. This means that with probability Q , participants are shown a correct decision, and with probability $1-Q$ participants are shown an incorrect decision. The quality is chosen randomly for each task: $Q \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ in two of the main experiments and $Q \in \{0.5, 0.9\}$ in the third. This signal provides participants with precise information about the quality of the decision that they could choose to learn from. Additional details on these signals and a discussion of the rationale behind structuring the signals in this manner are provided in Appendix C.2.

2.2 Conceptual framework

In these tasks, a decision maker (DM) must choose the correct decision y_T in order to earn an incentive. Although the DM has sufficient information to identify y_T , the cognitive demands of the task mean that a DM may find switching to the source's decision beneficial. In contrast, a fully Bayesian DM would identify the correct decision in the first period and would have no incentive to switch in the second period.

First, consider a context without identity. In this case, the DM's switching decision is a search for information quality and depends on their beliefs about the quality of the information provided by the source and their confidence in the accuracy of their initial decision. At the same time, DMs may care about more than accuracy – switching or sticking might involve psychological factors that would

influence DMs' choices. DMs switch if the utility from switching exceeds the utility from sticking:

$$U(y_2 = y_s, k_s) > U(y_2 = y_1)$$

where k_s represents the utility loss from switching, driven by factors such as conservative updating (Benjamin, 2019), resistance to new information (Weizsäcker, 2010; Conlon et al., 2022), or confirmation-seeking motives (Barron, 2021). Switching could also involve giving up possible psychological benefits from sticking, such as a preference for consistency (Falk and Zimmermann, 2018), or ownership effects (Hartzmark et al., 2021). While k_s could, in principle, be negative, theory and evidence suggest that it is likely to be positive.

In the experiment, the expected benefits from switching or sticking are directly affected by the provided quality signal ($Q = Pr[y_s = y_T]$) and the DM's confidence in their own decision's accuracy ($c = Pr[y_1 = y_T]$). If $k_s = 0$, then

$$y_2 = \begin{cases} y_s & \text{if } Q > c \\ y_1 & \text{if } Q \leq c \end{cases}$$

If $k_s > 0$, switching is costlier, meaning that such an agent would require a higher-quality source to switch than an agent with a lower (or even negative) k_s . This leads to the following testable predictions relating to beliefs: (i) switching increases with the quality signal Q , (ii) the more confident DMs are about the accuracy of their first decision (c), the less likely they are to switch to the source, and (iii) if $k_s > 0$, some agents would not switch when the quality of the source exceeds their confidence in the accuracy of y_1 .

Next, consider the case where the identity group g of the source is salient. Despite the lack of interaction or observability in this setting, identity can affect the switching decision through both beliefs and preferences.

Belief channel. The DM can infer the source's task-relevant abilities based on perceived characteristics of the group g , in a form of statistical discrimination (Phelps, 1972; Arrow, 1973) (or, if these beliefs are misspecified, inaccurate statistical discrimination (Bohren et al., 2025)). For illustration, suppose the DM combines the quality signal Q and a group-level prior Q_g as two signals of source quality, with variances σ_Q^2 and σ_g^2 . A Bayesian benchmark gives $\hat{Q} = \frac{\sigma_g^{-2}Q_g + \sigma_Q^{-2}Q}{\sigma_g^{-2} + \sigma_Q^{-2}}$, which shows that the precision of the experimentally provided signal directly affects the weight placed on it. In a setting where no quality signal is provided, DMs' beliefs about the quality of the information would be influenced by the identity g , meaning that DMs would switch more when the information source belongs to a group that (they believe) has higher ability (Q_g). In the main experiments, the quality signal Q is precise by design, meaning that no weight should be assigned to Q_g .

Preference channel. The DM may possess preferences over identity groups, which could lead the DM to ignore information not because of its quality but because of the source's identity, behaviour that would resemble taste-based discrimination (Becker, 1957; Guryan and Charles, 2013). Specifically,

Table 1: Experiment overview

Experiment	Identity Treatment	Sample	Sample size	Identity signal	Dates
<i>Main experiments:</i>					
Religion	<i>H</i> – Hindu	India panel	431	Surnames	April 2023
	<i>M</i> – Muslim		422		
Caste	<i>G</i> – General	India panel	415	Surnames	September 2023
	<i>O</i> – SC/ST		436		
Minimal	In-group	Prolific	142	Klee-Kandinsky	March 2023
	Out-group		136		

Notes. Participants are assigned to one experiment, and to one identity condition within that experiment. General – Source is from the General caste category. SC/ST – Source is from the Scheduled Castes or Scheduled Tribes categories. Appendix Tables B.2–B.4 present information on the demographics of the different samples.

if k_s varies based on the source’s identity group, it would imply that switching is costlier when the source belongs to an unfavoured group or out-group, and would be cheaper if the source belongs to a favoured group or in-group. This is motivated by social identity research (Tajfel et al., 1971; Akerlof and Kranton, 2000; Chen and Li, 2009; Shayo, 2009, 2020; Charness and Chen, 2020) which has documented that identity affects decision-making through identity-based behavioural prescriptions, self-image concerns, or a willingness to sacrifice material payoffs, all of which can affect decision-making even in the absence of observability or repeated association (as shown in Dekel and Shayo (2025), for example).

The main empirical challenge is that preferences for the identity of information sources and beliefs about group performance would jointly determine whether the DM chooses to switch. The goal of the experiment is to separate these two channels by experimentally assigning identity (between-subjects) and information quality (within-subjects). Under the identifying assumption that participants perceive that the signal precisely reveals the source’s true accuracy, the random assignment of the quality signal Q breaks the link between beliefs about information quality and identity. If the assumption holds, beliefs about the quality of the information provided by the source are unaffected by any identity-related beliefs, and observed differences in switching across treatment conditions that vary the information source’s identity can be attributed to preferences (or different costs) for learning from one identity group relative to another.

2.3 Main experiments

The three main experiments examine different identity contexts in which social identity has been shown to influence economic decision-making. Participants in an experiment are randomly assigned upon entry to one of two treatment conditions which differ only in the identity of the source. Table 1 lists the experiments in the study and Appendix Tables B.2–B.4 present summary statistics of the demographics of the different samples.

2.3.1 Experiment Caste

Background information. Caste is a system of social stratification in India that consists of thousands of caste groups called *jatis*. A person’s caste is largely determined by birth. Many prominent features of the caste system such as endogamy, social hierarchy, segregation, and ritual purity continue to influence modern Indian society (Mosse, 2020; Munshi, 2019). Caste groups are often associated with occupations (Cassan et al., 2021) – for example, some castes are associated with intellectual occupations, some with business, and others with farming. While the caste system is largely linked with Hinduism, non-Hindus may also hold caste identities. To effectively target welfare programs and affirmative action policies, the Government of India classifies castes into four “categories”. The “General” category (also known as the Forward Castes, $\approx 30\%$ of the population) consists of *jatis* that are considered socially and economically advanced (in relative terms). The “Scheduled Castes” (SC, $\approx 17\%$) and “Scheduled Tribes” (ST, $\approx 9\%$) categories consist of castes that are the most economically and socially disadvantaged. Last, the “Other Backward Classes” (OBC, $\approx 40\%$) category contains many *jatis* that are economically and socially disadvantaged relative to the General category.

Caste identity can influence both beliefs about information quality and preferences over the identity of information sources. First, individuals may hold beliefs about the task performance of people belonging to particular caste groups. This could be driven by economic realities such as disparities in education and income between caste groups (Asher et al., 2024), or by caste-specific stereotypes related to occupational and educational choices (Cassan et al., 2021; Oh, 2023). Second, people have strong preferences for associating with others from their caste groups or for avoiding members of caste groups considered lower in status (Munshi, 2019; Lowe, 2021). Such caste-dependent preferences could affect how people learn from sources belonging to different caste groups.

Implementation. The caste identity of the source is signalled through surnames that are informative of the caste category of the individual. These surnames are shown along with an arbitrarily chosen initial in place of “Study Participant” in Figure 1. There are two treatments in this experiment: *G* and *O*. In treatment *G*, surnames belonging to the General (or “Forward Castes”) category are used. In treatment *O*, surnames belonging to the Scheduled Castes or Scheduled Tribes are used. The surnames were validated for recognisability in a separate survey conducted through the same provider. A list of surnames and additional details are presented in Appendix C.3.

The *Caste* experiment and the validation study were conducted through online surveys with a gender-balanced sample of Indian Hindu participants recruited through an online panel survey provider. The effective sample size for the main experiment is 851 Hindu participants. Appendix Table B.2 presents descriptive statistics for the full sample and separately for each treatment. $\approx 64\%$ of the participants belonged to the General (*G*) caste, $\approx 34\%$ belonged to one of the other caste groups, and 2% did not report their caste group. The *G* category is over-represented in the sample relative to the general population. Participants in this experiment also responded to a set of questions on attitudes towards caste-based affirmative action, exposure to people from other castes/religions, and beliefs about the abilities of different caste/religious groups on the experimental task. Details are presented in Appendix C.4.

2.3.2 Experiment Religion

Background information. Religion is an important part of daily life in India.³ $\approx 80\%$ of the country is Hindu, $\approx 14\%$ is Muslim, and the rest of the population follows Christianity, Sikhism, Buddhism, Jainism, and other religions ([Census of India, 2011](#)). Religion moderates social and market interactions, and inter-religious relations are often characterised by a lack of trust. In particular, tensions between Hindus (the majority religion) and Muslims (the largest minority) have deep historical roots. Contemporary conflicts arise from a variety of factors such as territorial disputes, religious nationalism, political polarisation, and competition for resources ([Iyer, 2016](#); [Iyer and Shrivastava, 2018](#); [Jaffrelot, 2021](#)). [Pew Research Center \(2021\)](#) finds that $\approx 21\%$ of Muslims and 17% of Hindus report having experienced discrimination because of their religion in the past 12 months.

Muslims are, on average, poorer and less educated than Hindus ([Asher et al., 2024](#)), which may lead to beliefs that Muslims are less likely to be well-educated than upper-caste Hindus and therefore less likely to perform well on the balls-and-urns tasks. In terms of preferences, research has provided evidence of taste-based discrimination in the labour market ([Thorat and Attewell, 2007](#)), and of effects of religious identity on trust and pro-sociality ([Dhami et al., 2024](#)). Thus, it is plausible that religion can affect social learning through both belief and preference channels.

Implementation. The religious identity of the source is signalled using surnames that are informative of the religion of the individual. In treatment *Hindu*, the names are the same as those used in the *G* treatment of the *Caste* experiment. In treatment *Muslim*, the surnames are common Muslim surnames. Details on the surnames are presented in Appendix C.3.

This experiment was conducted through online surveys with a sample of participants from India recruited through a survey panel provider. The effective sample size is 853 participants, of whom $\approx 75\%$ were Hindus, $\approx 10\%$ Muslims, and the rest ($\approx 15\%$) were from other religions or did not disclose their religion. Muslims are under-represented in the sample relative to the population. Appendix Table B.3 presents descriptive statistics for the full sample and separately for each treatment.

2.3.3 Experiment Minimal

Background information. The minimal identity paradigm has been used for decades by social scientists to study the effects of identity on various decisions ([Tajfel et al., 1971](#); [Chen and Li, 2009](#)). Research has demonstrated that this method induces people to behave in a “groupy” manner by making choices that favour their in-group over their out-group ([Charness and Chen, 2020](#)), and has documented that identity affects learning: for example, [Berger et al. \(2018\)](#), [Parys and Ash \(2018\)](#), and [Zou and Xu \(2023\)](#) find that people learn more from their in-groups in experimentally assigned identity settings. This experiment sheds light on whether these findings are driven by beliefs or preferences. Additionally, as the minimal identity setting does not invoke considerations that arise with naturally occurring identities, this experiment provides a more controlled setting in which to examine these effects.

³Two-thirds of our sample say that religion is a very important part of their daily lives, and $\approx 90\%$ say that it is either somewhat or very important. This echoes findings from other surveys such as the Gallup World Poll and the World Values Survey.

Implementation. The experiment follows the method used in [Chen and Li \(2009\)](#) to assign minimal identities to participants. First, participants are asked to examine three pairs of paintings and indicate which they prefer in each pair.⁴ Next, participants are classified into either the Orange or Purple group based on whether they liked Klee (Purple) or Kandinsky (Orange) paintings. Participants are informed of the method by which the groups were assigned. Participants are also reminded of their assigned group identity once during the six tasks. Participants see sources as in Figure 1 with the addition of an orange or purple label with the text “Orange group” or “Purple group”, which depends on the participant’s group and the treatment to which they are assigned. In the *In-group* treatment, they see sources assigned to the same group, while in *Out-group* they see sources assigned to the other group.

Participants in this experiment were recruited through Prolific, a widely used survey platform for social science experiments, and came from the UK, EU, and the US. The effective sample size is 278 individuals. Appendix Table B.4 presents descriptive statistics for the full sample and separately for each treatment.

2.4 Supporting studies

Beliefs and surname recognition survey. This survey was conducted with 327 participants (effective sample of 267 Hindus) drawn from the same panel provider as the *Caste* experiments. The purpose of this survey was to gather incentivised beliefs about the performance of different identity groups on the experimental task, and to validate the chosen surnames for caste group recognisability.

Experiment *Caste* – No Signal. This pre-registered experiment was conducted concurrently with the *Caste* experiment. The only difference between these two experiments was that participants in *Caste* – No Signal did not receive a quality signal in the second period. The objective of this experiment was to study how people switch in the absence of a signal about the quality of information provided by the source. Appendix C.5 presents the details of this experiment.

Treatments *Computer* and *Continuous*. A pre-registered experiment was conducted concurrently with the minimal identity experiment. Participants were recruited through Prolific, and were assigned to one of three conditions: *Human*, *Computer*, or *Continuous*. In *Human*, sources are labelled as an anonymous “Study Participant”. In *Computer*, the source is labelled as a “Computer”. The comparison between *Human* and *Computer* provides a test of whether the experimental paradigm detects differences between source types that are more conspicuously differentiated than in the main experiments. This experiment is discussed in Section 4.1.3. The *Continuous* treatment is identical to the *Human* treatment, except that in the second period, participants use a slider to provide a posterior probability between 0 and 100, rather than the discrete switch decision. The aim of the *Continuous* treatment was to assess the extent of partial updating when participants are allowed to provide their own estimate in the second period.

⁴One of the paintings in each pair was created by the artist Paul Klee, and the other by the artist Wassily Kandinsky. Paintings by these artists have been used in many studies that use this method because of their similarity (at least, to the untrained eye).

2.5 Procedural details and design choices

Incentives. Participants complete six tasks and make two decisions in each task. One of the 12 decisions is randomly chosen for a bonus reward. Participants get a \$3 bonus payment if the chosen decision is within ± 2 percentage points of the true posterior; otherwise, they get no bonus. This incentive scheme was chosen both for simplicity and to avoid distorting participant decisions as highlighted in recent literature (Danz et al., 2022). All participants received a flat fee for participating in the study.

Data quality. To improve data quality, participants were allowed to proceed to the main tasks only if they passed two attention checks and correctly answered a battery of comprehension questions before any treatment information was revealed. Participants were given two opportunities to pass the test, as pre-registered. The pass rate in the *Religion* and *Caste* experiments was about 23%, and about 85% in the Prolific experiments.⁵ The low pass rates in the Indian samples limit generalisability to the broader Indian population. Appendix Tables B.2–B.4 show no evidence of systematic imbalance in demographic characteristics across treatment groups based on joint tests. The caste/religion compositions are similar across treatments in the *Caste* and *Religion* experiments.

Implementation. The experiments were programmed using oTree (Chen et al., 2016). The experiments on Prolific were conducted in March 2023. The *Religion* experiment was conducted in collaboration with Faktum Research in April 2023. The *Caste* experiments were conducted in September 2023 in collaboration with Norstat. The experiments were pre-registered at the AEA RCT registry (#0011066 and #0011924), and deviations are documented in Appendix C.9. All experiments were reviewed and approved by the IRB at the Norwegian School of Economics (NHH-IRB 39/22 and 44/22).

2.6 Descriptive Patterns

This section presents aggregate patterns in participants’ task behaviour in the experiments.

First-period performance. The bars in panel (a) of Figure 2 show the fraction of first-period decisions that are incentive-yielding (i.e. within ± 2 pp of y_T) in each experiment. The share of successful decisions ranges from $\approx 12\%$ in the Religion experiment to $\approx 14 - 15\%$ in the other experiments. First-period success rates do not vary significantly by participant identity in any of the experiments (panel A of Appendix Figures A.1 and A.2). Thus, while some participants make accurate decisions, most participants stand to improve their outcomes if they choose to switch in the second period. These patterns show that learning from others in this situation could be useful, fulfilling one of the design goals. The low success rates also reduce concerns that many participants mechanically computed the correct posterior probabilities (e.g. using spreadsheets or online tools), which could be a concern in the online experiment setting.

⁵Task comprehension for the balls-and-urns task is often low. For example, Enke and Graeber (2023) report a screening rate of above 50% in a Prolific sample on a belief updating task.

Second-period behaviour. The bars in panel (b) of Figure 2 show the fraction of decisions in which participants choose to learn by switching to the source in each experiment. The triangles in the plot represent the fraction of decisions in which switching would have been incentive-yielding (i.e., where the source was correct and the participant’s first-period decision was incorrect). The figure shows that while switching occurs in a considerable fraction of decisions (ranging from $\approx 24\%$ in the Religion experiment to $\approx 32\%$ in the Caste experiment), many participants leave money on the table by sticking with their own decision rather than switching to the source in the second period. Appendix Figure A.3 shows the distribution of the fraction of switching decisions in each experiment group. The figure shows that in all experiments, a small fraction of participants always switch, and that the modal participant never switches. The (b) panels of Appendix Figures A.1 and A.2 show that the switching behaviour of participants of different castes and religions is very similar, although Muslim and Christian participants switch slightly more than Hindu participants in the *Religion* experiment. An interpretation of these patterns is that participants find switching costly.

An alternative explanation for the low switching rates is that some participants may have preferred to update partially rather than completely switch to the source, which was not possible in the main experiments. Data from the supporting *Continuous* treatment, which elicits a continuous posterior, enable an assessment of the importance of this channel. Participants in this treatment demonstrate behaviour consistent with partial updating. For example, the fraction of decisions in which participants update by at least 5 percentage points relative to their initial estimate, and in which their second-period estimate remains at least 5 pp from the source is $\approx 23\%$. To compare rates of full updating, among decisions in which the first-period estimate was more than 2 pp away from the source, 26% of second-period estimates in *Continuous* were within 2 pp of the source, compared with a switching rate of 35% in *Human* in such decisions. Thus, while partial updating is important, behaviour resembling full updating occurs at nearly 75% of the switching rate.

In 2.1% of the decisions, participants’ first-period decisions were the same as the source’s decision. As the motive behind the second-period decision cannot be clearly attributed in such cases, these decisions are excluded from most of the analyses reported in this paper.⁶

3 Beliefs about information quality

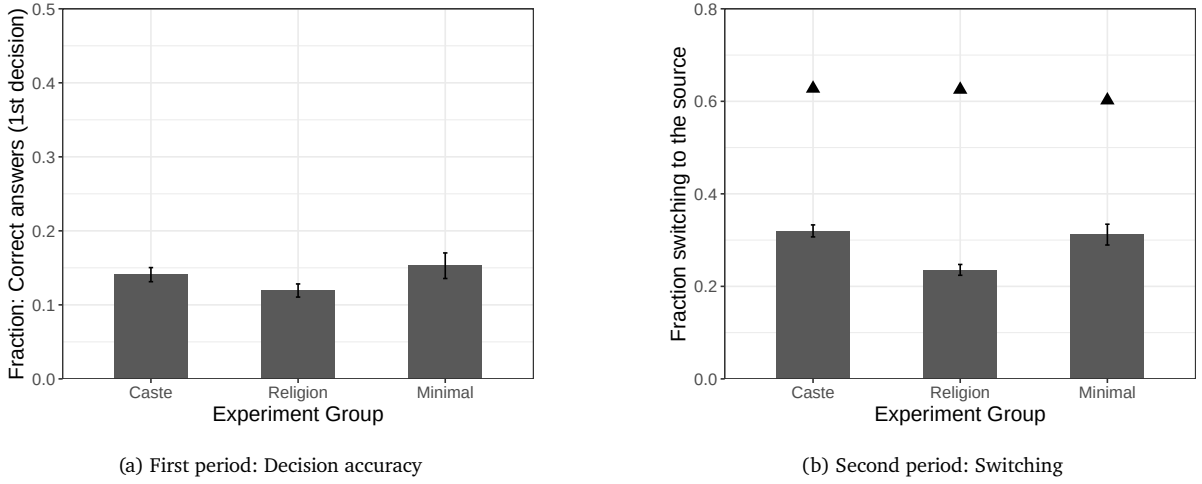
This section presents survey and experimental evidence on the effect of beliefs about information quality on learning when participants and information sources share the same information.

3.1 Descriptive evidence: Identity and performance beliefs

The beliefs survey, conducted with Indian participants from the same panel as the *Caste* experiments, elicited incentivised beliefs about the likelihood that a randomly chosen General Caste, Scheduled Caste, or Muslim individual would make an accurate first-period decision. To ensure that these beliefs were grounded in an understanding of the task, participants went through the same instructions and

⁶The results are qualitatively unaffected when these observations are included, or when the exclusion window is expanded to ± 1 (the pre-registered exclusion criteria), ± 2 , or ± 5 pp.

Figure 2: Descriptive patterns: Task performance and switching rates.



Notes. Panel (a): Bars indicate the fraction of decisions where participants made an incentive-yielding decision in the first period within each experiment. Panel (b): Bars indicate the fraction of decisions in which participants chose to switch to the source in the second period within each experiment. Triangles indicate the fraction of decisions where switching to y_s would have been incentive-yielding. Error bars indicate 95% confidence intervals.

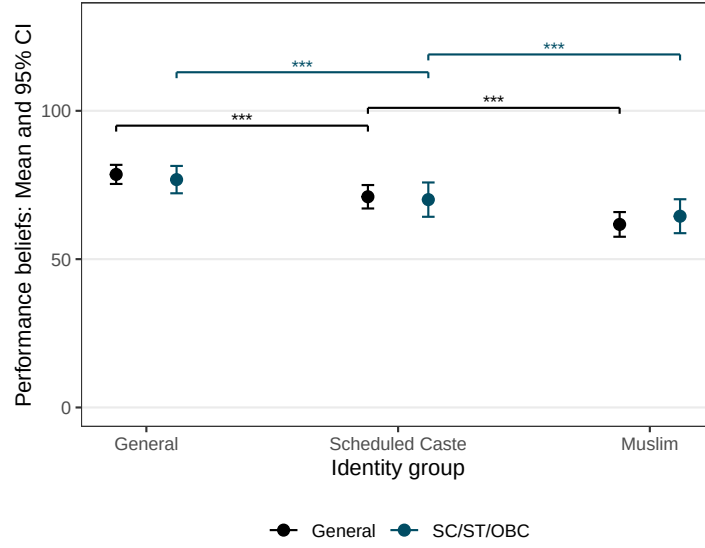
comprehension tests used in the main experiments before making the first-period decision on a single balls-and-urns task.

Figure 3 displays the means and 95% confidence intervals of these three beliefs separately for participants from the General Caste (G) and Hindus from the lower-status caste groups (O). The patterns indicate that participants believe that task performance differs by caste and religion. The average likelihood of success reported by G Hindus is 78.6% for fellow G Hindus, 71% for Scheduled Caste (SC) Hindus, and 61.7% for Muslims. The average likelihood of success reported by lower-status caste (O) Hindus is 76.8% for G Hindus, 70.1% for SC Hindus, and 64.5% for Muslims. Hypothesis tests indicate that Hindus in our sample believe that G Hindus outperform SC Hindus, and that both General and SC Hindus outperform Muslims. Panels (a) and (b) of Appendix Figure A.4, which show the cumulative distributions of these beliefs, demonstrate that the average differences are visible across most of the distribution. The p-values from Kolmogorov-Smirnov tests for the equality of distributions are < 0.01 in both cases.⁷

Figure 3 also shows that participants from different caste groups hold similar beliefs about the likelihood of success of the different groups. At the same time, these beliefs are poorly calibrated: participants overestimate the performance of all identity groups, given that the actual success rates are $\approx 12 - 15\%$ (see Figure 2). Together, these patterns provide descriptive evidence that using identity to infer information quality in the context of the experimental task could be misleading because performance beliefs are, on average, optimistic, and perceived differences between groups do not match actual performance differences.

⁷Beliefs about the ability of different caste groups were also elicited (post-treatment) in the *Caste* and *Caste – No Signal* experiments. Appendix Figure A.5 shows that the patterns are similar when using post-treatment beliefs from those experiments.

Figure 3: Beliefs about the performance of different groups



Notes. The figure shows mean incentivised beliefs about the likelihood that a randomly chosen person from each identity group would make an accurate first-period estimate in the task, separately for General caste (black dots) and SC/ST/OBC Hindu (blue dots) participants. Each pair of estimates corresponds to the subject of the elicited performance belief. Error bars indicate 95% confidence intervals. Stars above brackets indicate significance levels for paired comparisons, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

3.2 Experimental evidence: Beliefs about information quality

Participants in the main experiments receive a signal about the quality of the information provided by the source, which is the probability that the shown decision is correct. The quality signal is randomly assigned for each task and varies between 50% and 90%. Figure 4 shows the fraction of decisions in which participants switch at each level of the quality signal, separately for each of the experiments. As the quality of information increases from 50% to 90%, switching in the *Caste* and *Religion* experiments increases by $\approx 6 - 8$ percentage points and by ≈ 25 percentage points in the *Minimal* experiment.

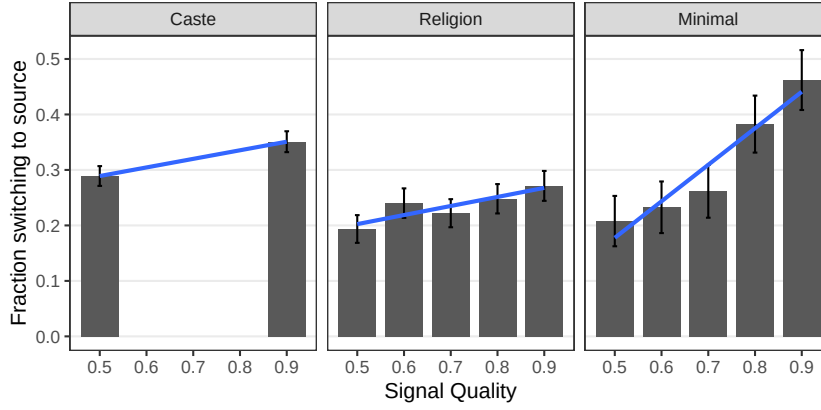
Table 2 presents results from regressions of the following specification for each experiment:

$$\text{Switch}_{i,n} = \beta_0 + \beta_1 Q_{i,n} + \beta_2 M_{i,n} + X_i + T_i + v_n + \epsilon_{i,n}, \quad (1)$$

where $\text{Switch}_{i,n} = 1$ if participant i switches to the source in the second period of task n and 0 otherwise. $Q_{i,n}$ is the level of the quality signal seen by participant i in task n . T_i is an indicator for the treatment that participants are randomly assigned to: $T_i = 1$ if the participant is assigned to see a General Caste source (in *Caste*), a Hindu source (*Religion*), or an in-group source (*Minimal*).⁸ $M_{i,n}$ is a set of variables including the participant's confidence in their first-period decision, beliefs about the performance of the assigned source's caste group (in the *Caste* experiments), an indicator for whether the source's shown decision was correct, the absolute difference between the participant's first decision and the source's decision (discussed in Appendix C.7), and an indicator variable

⁸The analysis of the treatment differences is presented in Section 4.

Figure 4: Effect of beliefs about information quality on learning



Notes. The graph shows the fraction of decisions in which participants choose to switch to the source in each experiment, at each level of source quality Q . $Q \in \{.5, .9\}$ in *Caste*, and $Q \in \{.5, .6, .7, .8, .9\}$ in *Religion* and *Minimal*. Error bars indicate 95% confidence intervals.

for whether their first guess was correct. X_i is a set of demographic controls (age, gender, tertiary educational attainment, and employment status). v_n includes order and task fixed effects. Standard errors are heteroscedasticity-robust and clustered at the participant level. The coefficient of interest is β_1 , which identifies the causal effect of an increase in the quality of information on switching. β_2 captures the association between switching and other variables of interest.

Columns (i)–(vi) of Table 2 show that the relationship between the quality signal and switching is positive and statistically significant in each of the three main experiments – participants are more likely to switch when quality is higher. These results can be given a causal interpretation as the quality signal is exogenously assigned at the task level. The responsiveness to an increase in quality varies across samples and is more pronounced in experiment *Minimal*. Figure 4 also shows that switching is relatively low even when information quality is high. When the quality of the shown information is 90%, switching ranges from approximately 27% to 46% in the different experiments. These patterns suggest that participants find switching costly, and are consistent with prior research documenting that people do not learn from others even though they would benefit by doing so (Weizsäcker, 2010; Benjamin, 2019; Conlon et al., 2022).

Group performance beliefs and switching. Participants in the *Caste* and *Caste – No Signal* experiments also stated incentivised beliefs about the likelihood of success of a random person belonging to the General or SC caste group. These beliefs were elicited after the main tasks using the same question as in the separate beliefs survey discussed above. The estimates in Columns (ii) and (viii) of Table 2 include the relevant belief variable (i.e. the belief about the likelihood of success of the caste group corresponding to their assigned identity condition). The estimates show that while these beliefs are positively associated with switching in the *Caste – No Signal* experiment (Column (viii)), there is no significant association in the *Caste* experiment (Column (ii)). Consistent with the quality signal not being shown in the *Caste – No Signal* experiment, the coefficient on assigned but undisclosed source quality for this experiment (Columns (vii–viii)) is smaller in magnitude and not statistically distinguishable from 0.

Table 2: Regression analysis: Beliefs and switching.

	Dep. var.: Switch to source							
	Caste		Religion		Minimal		Caste (No Signal)	
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Quality	0.154*** (0.036)	0.138*** (0.038)	0.163*** (0.043)	0.124*** (0.045)	0.665*** (0.086)	0.615*** (0.090)	0.074 (0.057)	0.047 (0.059)
Beliefs (Identity group)		0.080 (0.050)						0.228*** (0.076)
Confidence (c)		-0.319*** (0.041)		-0.227*** (0.037)		-0.436*** (0.055)		-0.237*** (0.062)
Guess 1 correct ($ y_1 - y_T \leq 2$)		-0.102*** (0.021)		-0.043** (0.019)		-0.116*** (0.028)		-0.067** (0.034)
Constant	0.226*** (0.029)	0.378*** (0.070)	0.126*** (0.033)	0.331*** (0.063)	-0.144** (0.062)	0.186* (0.096)	0.234*** (0.047)	0.324*** (0.114)
Task controls		✓		✓		✓		✓
Demog. controls		✓		✓		✓		✓
Treatment FE	✓	✓	✓	✓	✓	✓	✓	✓
R ²	0.005	0.041	0.003	0.037	0.041	0.127	0.002	0.047
Dependent variable mean	0.320	0.320	0.236	0.236	0.312	0.312	0.273	0.273
Observations	4,952	4,952	5,063	5,063	1,629	1,629	1,719	1,719
Individuals	851	851	853	853	278	278	295	295

Notes. Estimates from OLS regressions where the dependent variable is whether a participant switches in the second period. The main variable of interest is *Quality*, which is the experimentally assigned quality signal (not displayed in the *No signal* experiment). Other variables: (i) beliefs about the likelihood of success of the identity group in the assigned treatment (in *O*, beliefs about the performance of Scheduled Castes are used as a proxy for beliefs about the *O* source), (ii) participant's reported confidence in their first-period decision (0 to 1), and (iii) whether y_1 was accurate (0/1). Controls are task and order fixed effects, source correctness, disagreement, demographic characteristics (the participant's age and dummies for employment, college education, and gender), and controls for the identity treatment within each experiment. Robust standard errors, clustered at the participant level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

For comparison, a 40 percentage point difference in beliefs about the identity group (equivalent to the experimental variation in the quality signal) would imply an increase of ≈ 9 pp in switching, against the ≈ 6 pp increase observed in the *Caste* experiment. These patterns suggest that when information quality is unknown, participants may rely on underlying beliefs about group-specific performance to gauge information quality when deciding whether to switch. However, as beliefs are elicited after the main tasks, these patterns should be interpreted as associations rather than causal effects.

Confidence and switching behaviour. The balls-and-urns setup enables an investigation of how confidence, a key behavioural mechanism, influences learning in the experiment. Participants' reluctance to switch to the source may be partially driven by overconfidence in the accuracy of their first-period decision. The median reported confidence that the first-period decision will earn the incentive is $\approx 85\%$ in the *Caste* and *Religion* experiments and $\approx 70\%$ in the *Minimal* experiment. These levels are much higher than the realised accuracy of first-period decisions. Participants behave in line with these beliefs: the estimated coefficient on confidence is negative and statistically significant in all experiments, meaning that higher confidence in one's first-period decision is linked with a lower rate of switching. Appendix Figure A.6 shows this visually – Panel (a) shows that more confident people make smaller errors (measured by the absolute difference between y_1 and y_T), and

Panel (b) shows that more confident people are less likely to switch in the second period. Participants who make accurate first-period decisions are also much less likely to switch than participants who fail to make an accurate decision. Thus, while higher confidence is somewhat justifiable given the correlation with smaller errors, the low accuracy of first-period decisions (12 – 15%) suggests that participants are overconfident. Appendix Figure A.7 supports this interpretation, showing that participants switch more when information quality is at least as high as their confidence in their own decision. The consistency of these patterns across all of the experiments indicates that participants' beliefs about their own performance are an important correlate of the switching decision.

Summary. These results provide evidence that (i) information about source quality causally influences switching, (ii) learning rates are low, even when information quality is high, and (iii) switching is systematically related to both source quality and subjective confidence.

4 Preferences for information source identity

This section examines whether participants respond differentially to source identity when information quality is experimentally provided in the three main experiments. It begins with an analysis of the average treatment effects of the between-subjects identity assignment on switching behaviour, moves on to an exploration of heterogeneity and alternative explanations for the main results, and concludes with an analysis of in-group preferences in the *Caste* and *Religion* experiments.

I estimate the following specification using OLS regressions:

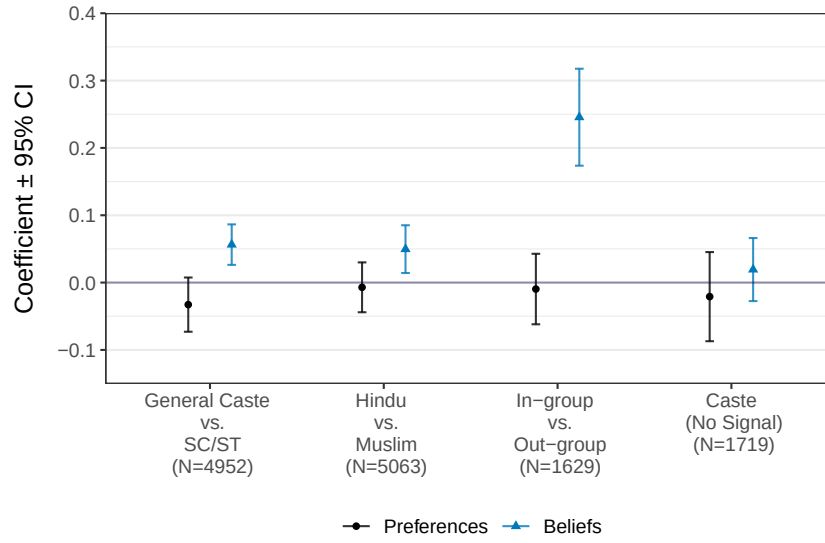
$$Switch_{i,n} = \beta_0 + \beta_1 T_i + \beta_2 Q_{i,n} + M_{i,n} + X_i + v_n + \epsilon_{i,n}, \quad (2)$$

where *Switch* is an indicator variable for participant *i* in task *n*. *Switch* = 1 if the participant switches to the source in the second period and 0 otherwise. *T_i* is an indicator for the treatment group that participants are randomly assigned to: *T_i* = 1 if the participant is assigned to see a source who belongs to the General Caste (in *Caste*), is a Hindu (*Religion*), or is from their In-group (*Minimal*). *Q_{i,n}* is the level of the quality signal seen by participant *i* in task *n*. *X_i* is a set of demographic controls (age, gender, college education, and employment status). *M* is a set of additional controls such as *i*'s confidence in the accuracy of their first-period decision and whether the shown decision is correct. *v_n* is a vector of task-specific controls. Standard errors are heteroscedasticity-robust, clustered at the participant level. The coefficient of interest is β_1 , which under the identifying assumption can be interpreted as an estimate of preferences for learning from one identity group over another when the quality of information is known.

Hypothesis 1. *The estimate of $\beta_1 \neq 0$ in each of the experiments, indicating that participants prefer to switch more to sources from one of the two identity groups: (i) a General caste or SC/ST caste, (ii) a Hindu or a Muslim, (iii) experimentally assigned in-group or out-group.*

Figure 5 presents the results from estimating this equation separately for each of the main experiments and the *Caste – No Signal* experiment (Appendix Tables B.5–B.8 present the corresponding

Figure 5: Preferences for information source identity: Average treatment effects



Notes. Each dot is the coefficient on the treatment indicator from OLS regressions estimating Equation 2 within each experiment (reported in Column (vi) of Appendix Tables B.5–B.8); N is the number of observations. Triangles show the rescaled coefficient on signal quality (a 40 pp increase) from the same set of estimates for a visual comparison. The dependent variable is whether a participant switches in the second period. Controls are task and order fixed effects, whether the source is correct, first-period confidence, and demographic characteristics (the participant’s age and dummies for employment, college education and gender). Bars indicate 95% confidence intervals based on robust standard errors, clustered at the participant level.

regressions). The black dots and error bars in the figure show the estimated coefficient and 95% confidence intervals for the treatment indicator. The average treatment effects in the *Caste*, *Religion*, and *Minimal* experiments are small and statistically indistinguishable from zero. There is no evidence to support Hypothesis 1 that participants have different preferences for switching to a *G* caste or *O* caste source in the *Caste* experiment (-3.3 pp, $p = 0.11$, 95% CI: $[-7.3, 0.8]$), a Hindu or a Muslim source in the *Religion* experiment (-0.7 pp, $p = 0.71$, 95% CI: $[-4.4, 3.0]$), or an (experimentally assigned) in-group or out-group source in the *Minimal* experiment (-1.0 pp, $p = 0.72$, 95% CI: $[-6.2, 4.3]$). The 95% confidence intervals are within 0.2 standard deviations of the dependent variable in the main experiments.

The triangles in Figure 5 represent the estimates for beliefs (in blue) from the same regressions. To provide an economically meaningful comparison, the coefficient on quality is rescaled to represent a 40 pp difference in quality. The two sets of estimates provide a visual comparison of the relative magnitudes of the belief and preference channels – while the quality signal influences switching across the main experiments, the source identity effects are small and statistically indistinguishable from zero.

The statistically insignificant treatment effects may reflect true null effects, or a lack of statistical power to detect small effect sizes. Appendix Table B.9 reports the estimates in percentage points, together with confidence intervals, and equivalence tests. Following the two one-sided tests (TOST) procedure (Lakens et al., 2018), the tests support equivalence within ± 10 pp in all three main experiments. The null result is more informative in the *Religion* experiment, where an absolute effect

size of 5 pp can be ruled out. Thus, while the null results on preferences are informative about moderate-to-large effects, they cannot rule out smaller effect sizes.

Do preferences emerge without a quality signal? The fourth set of estimates in Figure 5 is for experiment *Caste – No Signal*. The estimated treatment effect for caste identity in this experiment is not statistically significant, which does not provide evidence of differences in switching because of the source’s caste identity when the quality of the source is not experimentally provided. This result stands in contrast with the results on beliefs in Section 3, which showed that there are differences in participants’ beliefs about the performance of different identity groups. To place these results in perspective, note that in the *Caste* experiment a 40 pp (exogenous) increase in source quality leads to a ≈ 6 pp increase in switching. In comparison, the average believed difference in ability between the two caste groups is much smaller, ≈ 7 pp, which translates to an effect size of about 1 pp. Thus, if there are no strong preferences for the caste identity of information sources, the *No Signal* experiment (with a sample size of 295 participants) is underpowered to detect small differences in switching because of the difference in beliefs about the performance of different groups.

4.1 Robustness and heterogeneity

The detailed regression analyses presented in Appendix Tables B.5–B.8 show that the estimated average treatment effects are not sensitive to demographic characteristics, task-specific controls, or contextual features that affect belief updating such as subjective confidence or source disagreement. Appendix Table B.10 shows that the results in the *Religion* experiment are not driven by participants who are neither Hindu nor Muslim.

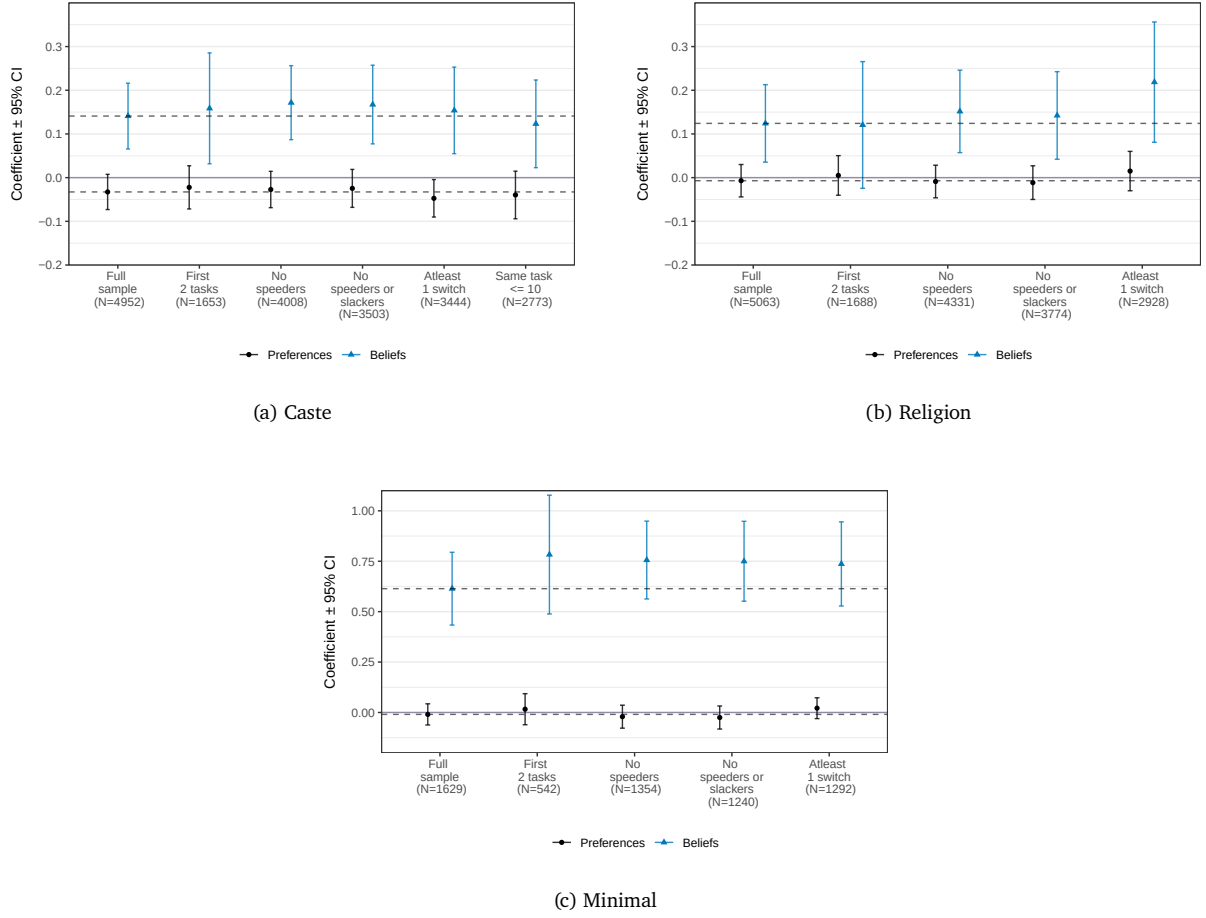
However, the average treatment effects may be affected by factors such as inattention, fatigue, or imperfect comprehension, and could mask heterogeneity in identity preferences. This part of the analysis investigates whether treatment effects vary across theoretically relevant subgroups, focusing on task engagement and well-documented behavioural mechanisms. Diagnosing differences along these dimensions is important as the identifying assumption requires that participants understand the information structure, and especially the quality signal.

4.1.1 Comprehension, fatigue, or inattention

Figure 6 shows the results from a battery of diagnostic tests examining whether the results are driven by a lack of comprehension of the task, confusion, or inattention – plausible concerns given the cognitively demanding nature of the balls-and-urns task and the online setting. These tests use features of participants’ decisions that could represent random decision-making, cognitive fatigue, or a lack of understanding of the task and incentive structure. The panels show estimates of the average treatment effect (circles) and the effect of signal quality (triangles) for each experiment under different sample restrictions, all using the specification in Equation 2.

First, participants may be more motivated or likely to remember instructions at the beginning of the experiment. Restricting the sample to the first two (of six) tasks results in more imprecise estimates, but does not appear to qualitatively affect the results. Next, completing tasks very quickly

Figure 6: Robustness tests: beliefs and preferences



Notes. Circles represent the estimated coefficient on the treatment dummy variable (Preferences) from OLS regressions of Equation 2 under different sample restrictions (N is the number of observations). Triangles represent the (unscaled) coefficient on signal quality (Beliefs) from the same regressions. Error bars indicate 95% confidence intervals. Standard errors are clustered at the participant level. Controls: demographics, task and order fixed effects, confidence, whether the source was correct, and the quality of the information source. The dashed lines represent the full-sample estimates. Panel (c) uses different y-axis bounds as the coefficient on quality is larger in *Minimal*.

or very slowly may reflect inattention or distraction. The third and fourth estimates in each panel exclude decisions that are roughly in the bottom (and top) 10% of time taken while making the switching decision. Excluding such decisions has virtually no effect on the estimates. Next, participants who never switch may arguably do so because they just want to get through the tasks quickly. When the sample is restricted to participants who switch at least once, the estimates are very similar in two of the three experiments. The estimated treatment effect (for preferences) is statistically significant and negative at the 5% level in the *Caste* experiment. Finally, in experiment *Caste*, the last of the six tasks was the same as one of the first two tasks.⁹ The estimate on a restricted subsample of participants whose decisions on identical tasks differ by ≤ 10 pp – an indicator of high consistency – is very similar to the estimates using the full sample.

⁹This design feature was included to specifically test for random decision-making. The correlation between participants' responses on the identical tasks is 0.63.

4.1.2 Task features and decision moderators

Next, I examine the interaction between factors that affect switching and the identity of the information source. These factors include experimentally controlled parameters and endogenous responses measured during the task: (i) experimental variation in the quality of the information (whether responsiveness to quality varies by source identity), (ii) experimental variation in whether the source is correct or incorrect (whether participants discern incorrect sources differently based on identity), (iii) the participant’s confidence in their first-period decision (whether the quality-confidence trade-off varies by identity), (iv) the accuracy of the participant’s first-period decision (whether high or low performing participants react differently to source identity), (v) the difference between the first-period decision and the source’s decision ($|y_1 - y_s|$, whether the sensitivity of switching to source disagreement differs by identity), and (vi) the time spent when making the switching decision (whether more deliberative decisions differ by source identity). Conceptually, evidence that these factors affect behaviour based on the source’s identity would support the existence of conditional identity preferences.

I estimate Equation 2, adding an interaction term between a factor and the treatment indicator. Appendix Tables B.11, B.12, and B.13 present the results from estimating this specification for each of the six mechanisms within each experiment. Across experiments, the results provide very little evidence that these mechanisms interact with identity. Across the 18 different interactions reported above, only the estimated coefficient on the interaction term in the *Religion* experiment (Column (i) of Table B.12) is statistically significant at the 5% level. This estimate suggests that participants may respond less to an increase in the quality of a Hindu source than to an equivalent increase for a Muslim source. While this result suggests that source identity may affect the interpretation of the assigned quality signal in the case of religion, it does not survive after correcting for multiple hypotheses.

Other sources of heterogeneity. Appendix C.8 presents a summary of pre-registered heterogeneity analyses of exposure to individuals from different castes/religions (in the *Caste* experiment), and individualism (in the *Religion* experiment). These analyses provide no evidence of heterogeneous preferences along these dimensions.

4.1.3 Human vs. computer sources

A possible interpretation of these results is that the experimental paradigm may not be equipped to identify differences in switching because of source characteristics. The *Computer* experiment was designed to test whether the paradigm could generate a difference in switching in a conceptually related context. It examines a stronger distinction – whether social learning (where the source is another human) differs from non-social learning (where the source is a computer algorithm). The experiment consists of two treatments: *Human* (in which sources are labelled as “Study Participant”, as in Figure 1), and *Computer* (in which the source is labelled as “Computer” and the human icon is replaced with a computer icon). The *Human* treatment uses the same structure as the main experiments, while the *Computer* treatment frames the switching decision as a lottery (without using the word). Appendix C.6 contains implementation details and further discussion about this experiment.

Appendix Table B.14 presents estimates of Equation 2 for this experiment. The estimates show that participants are substantially more likely to switch when the source is labelled as a computer rather than as another human. The estimated effect is about 15 percentage points and is robust to the inclusion of task controls, demographic controls, source correctness, quality, and first-period confidence. Appendix Figure A.8 shows that this difference is present across levels of source quality, and Appendix Table B.15 suggests that source disagreement is associated with a smaller difference in switching between human and computer sources.¹⁰

These results show that the experimental paradigm can generate large, precisely estimated, and robust differences in switching in response to changes in source presentation and framing, reducing the possibility that the null results in the main experiments reflect a general insensitivity of either the paradigm or the switching outcome to source representation.

4.2 In-group preferences and overlapping identities

A person’s identity *relative* to that of the information source could also influence switching in the experiment, given previous research showing that identities affect decision-making through in-group favouritism and out-group parochialism (Shayo, 2020; Charness and Chen, 2020). While the *Minimal* experiment finds no evidence of in-group preferences with experimentally assigned identities, the *Religion* and *Caste* experiments allow for further investigation with naturally occurring identities. Participants in these experiments self-report their caste group and religion, enabling an analysis of heterogeneity in preferences for information from an in-group (same caste category or religion) rather than an out-group source (different caste category or religion).

I estimate the following specification using OLS regressions:

$$\text{Switch}_{i,n} = \beta_0 + \beta_1 T_i + \beta_2 I_i + \beta_3 T_i \times I_i + Q_{i,n} + M_{i,n} + X_i + v_n + \epsilon_{i,n}, \quad (3)$$

where I_i is an indicator variable for the participant’s identity: in experiment *Caste*, $I_i = 1$ if the participant belongs to the general category (*G*) and $I_i = 0$ if the participant belongs to the SC/ST/OBC categories. In experiment *Religion*, $I_i = 1$ if the participant is Hindu and $I_i = 0$ if the participant is Muslim. The sample for the *Religion* experiment is restricted to Hindu and Muslim participants to allow for the construction of unambiguous in-groups. Participants who do not report a caste group are also excluded from these analyses (14 in *Religion* and 17 in *Caste*).

The coefficient β_1 is the difference in switching when seeing a $T_i = 1$ source relative to a $T_i = 0$ source for $I_i = 0$ participants. $\beta_1 + \beta_3$ is the difference in switching when seeing a $T_i = 1$ source relative to a $T_i = 0$ source for $I_i = 1$ participants. The interaction coefficient β_3 is the difference in these differences between $I_i = 0$ and $I_i = 1$ participants.

¹⁰Overall, these results are consistent with a stronger preference for non-human sources, or “algorithm appreciation” (Logg et al., 2019). The evidence on whether people learn more from humans or computers is mixed. While some find that people are more averse to learning from algorithms (Goeree and Yariv, 2015; Dietvorst et al., 2018, 2015), others find evidence of algorithm appreciation (Logg et al., 2019). Hou and Jung (2021) show that appreciation or aversion emerges because of differences in framing and perceived accuracy, and Langer et al. (2022) document the sensitivity of people’s responses to the terminology used to refer to the non-social source.

Hypothesis 2. *The estimate of $\beta_3 \neq 0$, indicating that participants learn differently from an in-group source than from an out-group source within the Religion and Caste experiments.*

Columns (i)-(ii) of Table 3 present the results from estimating Equation 3 in the *Caste* and *Religion* experiment samples. None of the estimated coefficients of interest are statistically significant. Neither is there any evidence that participants belonging to different castes or religions switch differently, nor are the treatment effects for any of the subgroups statistically significant. Taken together, there is no evidence that participants belonging to different castes or religions have in-group preferences for information from sources that share their caste group or religion.

These results are robust to the inclusion of task and demographic controls, and to controlling for participants' stated confidence. Appendix Tables B.16 and B.17 present the results from the same diagnostic tests used to evaluate the average treatment effects. The tables show that the estimates are largely consistent with the estimates in Table 3.

Overlapping identities. The analysis so far has focused on a single dimension of identity. However, in naturally occurring contexts, people possess multiple identities which may affect learning in different ways. In the Indian context, people have both religious and caste identities, and the importance and manifestation of behavioural norms and prescriptions associated with caste identity often vary, both between caste groups and across religions (Munshi, 2019; Mosse, 2018, 2020). The *Religion* experiment includes participants of different religions who also hold caste identities, which makes it possible to explore how preferences for the religious identity of sources vary across different intersections of religious and caste identities.

I estimate the following specification to study the effect of overlapping religious and caste identities in the *Religion* experiment:

$$\text{Switch}_{i,n} = \beta_0 + \beta_1 T_i + \beta_2 C_i + \beta_3 T_i \times C_i + Q_{i,n} + M_{i,n} + X_i + v_n + \epsilon_{i,n}, \quad (4)$$

where C_i is the caste identity of participant i . $C_i = 0$ if the participant belongs to the SC/ST/OBC caste category, and $C_i = 1$ if the participant belongs to the general caste category.

Columns (iii)–(iv) of Table 3 present estimates of Equation 4 using subsamples of Hindu and Muslim participants within experiment *Religion*. Column (iii) shows that Hindu participants belonging to the SC/ST/OBC castes switch about 9 pp less ($p = 0.018$) when they see a Hindu source than when they see a Muslim source. The interaction term (Hindu Source $\times$ General Caste) is positive and statistically significant ($p = 0.006$), indicating that the switching difference (between Hindu and Muslim sources) is higher for General caste participants than for SC/ST/OBC participants. Turning to the Muslim subsample (Column (iv)), there is no evidence that switching to Hindu or Muslim sources differs by the (Muslim) participant's caste group. However, the subsample of Muslims is substantially smaller than the Hindu subsample.

These results provide exploratory evidence that preferences for source identity may vary across different intersections of caste and religious identity. The analysis also highlights the importance of accounting for the fact that individuals belong to many identity groups concurrently.

Table 3: Regression analysis: Heterogeneous effects by caste and religion

	<i>Dep. var.: Switch to source</i>			
	In-group Preferences Experiment		Overlapping Identities Participant Religion	
	Caste	Religion	Hindu	Muslim
	(i)	(ii)	(iii)	(iv)
General Caste Source	-0.015 (0.036)			
General Caste Source $\times$ General Caste	-0.030 (0.044)			
Hindu Source		0.068 (0.065)	-0.092** (0.039)	0.055 (0.087)
Hindu Source $\times$ Hindu		-0.076 (0.068)		
Hindu Source $\times$ General Caste			0.128*** (0.047)	0.018 (0.123)
General Caste	-0.003 (0.032)		-0.093*** (0.036)	0.003 (0.078)
Hindu		-0.002 (0.042)		
Quality	0.144*** (0.039)	0.080 (0.049)	0.092* (0.051)	0.029 (0.168)
Constant	0.407*** (0.068)	0.375*** (0.079)	0.414*** (0.078)	0.339 (0.208)
Task controls	✓	✓	✓	✓
Demog. controls	✓	✓	✓	✓
R^2	0.034	0.024	0.029	0.068
Dependent variable mean	0.320	0.228	0.224	0.266
Observations	4,852	4,264	3,775	489
Individuals	834	719	636	83

Notes. Columns (i)–(ii): Estimates of Equation 3 for in-group preferences in the *Caste* and *Religion* experiments. Columns (iii)–(iv): Estimates of Equation 4 for in-group preferences with overlapping identities using subsamples of Hindu and Muslim participants in the *Religion* experiment. The dependent variable is 1 if the participant switches to the source’s decision y_s . “General Caste Source” and “Hindu Source” indicate whether a participant was assigned to see that source, and “General Caste” and “Hindu” indicate whether a participant belonged to the General caste or was a Hindu. Demographic controls are the participant’s age and dummies for employment, college education, and gender. Task controls include task and order effects, first-decision confidence, and an indicator for whether the source is correct. Standard errors are clustered at the participant level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Conclusion

This paper studies how the identity of an information source affects whether people learn, using online experiments with naturally occurring and experimentally induced identities. The main contribution of this paper is to independently vary information quality and source identity to show that while participants respond to an experimentally assigned signal of source quality, there is little evidence of residual preferences for the identity of the information source conditional on receiving the quality signal. These results complement recent work showing that identity can affect information acquisition and belief formation (Robbett et al., 2023; Zhang and Rand, 2023; Dekel and Shayo, 2025; Bauer et al., 2026; Gangadharan et al., 2025) by identifying a boundary condition: identity has little

residual influence on learning when quality is known, even though participants report substantial perceived performance differences between identity groups. The absence of systematic evidence of taste-based preferences in this learning environment is also an encouraging result, especially given the large literature on caste and religious identity in India which documents taste-based discrimination in a variety of decision-making settings.

The results should be interpreted in light of the experimental framework and implementation context. The *Caste* and *Religion* experiments use Indian online panel samples and therefore may not capture the full heterogeneity of the broader Indian population. The abstract balls-and-urns task allows the isolation of the role of source identity by providing information about source quality and limiting signalling and social interaction. These features may interact with source identity in richer settings, which the present design does not examine. For example, identity may matter differently when information is embedded in social relationships or closely tied to identity concerns, as with policy information in U.S. partisan settings (Robbett et al., 2023; Bauer et al., 2026). The experimental paradigm developed in this paper could be adapted to study the relative roles of beliefs and preferences in richer settings involving repeated interactions, identity-relevant information, and multiple intersecting identities.

Despite these limitations, the results carry implications for policy-makers and organisations. Providing a strong signal of information quality, while challenging in many policy communication settings, could reduce belief-driven distortions in information weighting. Future work could develop and test methods to signal information quality with precision and credibility. Beyond identity concerns, such research would also speak to ongoing programs on combating disinformation, changing social norms, and increasing the adoption of welfare programs.

Additional Information

Acknowledgements. I gratefully acknowledge guidance and support from Bertil Tungodden, Erik Ø. Sørensen, and Benjamin Enke. Kata Urban provided excellent administrative support. I further thank Kai Barron, Alexander W. Cappelen, Daniel Carvajal, Fabio Galeotti, Holger Herz, Sam Hirshman, Sissel Jensen, Michael Kosfeld, Julien Senn, and participants at various seminars and conferences for valuable comments.

Funding source. This study was supported by grant 788433 from the European Research Council.

Pre-registration and ethics review. The experiments were pre-registered at the AEA RCT registry (#0011066 and #0011924). All experiments were reviewed and approved by the Norwegian School of Economics Institutional Review Board (NHH-IRB 39/22 and 44/22).

AI usage disclosure. Commercial AI tools (ChatGPT and Codex by OpenAI) were used for (i) proofreading the text, (ii) review and optimisation of R code written by the author, and (iii) literature search and bibliography audit. Any modifications based on output from these tools have been verified before incorporation. The author takes full responsibility for all content.

References

- Afrouzi, Hassan, Carolina Arteaga, and Emily Weisburst**, “Is It the Message or the Messenger? Examining Movement in Immigration Beliefs,” *Journal of Political Economy Microeconomics*, May 2024, 2 (2), 244–297.
- Agranov, Marina, Gabriel Lopez-Moctezuma, Philipp Strack, and Omer Tamuz**, “Learning through Imitation: An Experiment,” *Journal of Economic Theory*, 2026, 235, 106203.
- Akerlof, George A. and Rachel E. Kranton**, “Economics and Identity,” *Quarterly Journal of Economics*, aug 2000, 115 (3), 715–753.
- Arrow, Kenneth J.**, “The Theory of Discrimination,” in Orley Ashenfelter and Albert Rees, eds., *Discrimination in Labor Markets*, Princeton, NJ: Princeton University Press, 1973, pp. 3–33.
- Asher, Sam, Paul Novosad, and Charlie Rafkin**, “Intergenerational Mobility in India: New Measures and Estimates across Time and Social Groups,” *American Economic Journal: Applied Economics*, April 2024, 16 (2), 66–98.
- Atkin, David, Eve Colson-Sihra, and Moses Shayo**, “How Do We Choose Our Identity? A Revealed Preference Approach Using Food Consumption,” *Journal of Political Economy*, apr 2021, 129 (4), 1193–1251.
- Banerjee, Abhijit, Esther Duflo, Maitreesh Ghatak, and Jeanne Lafortune**, “Marry for What? Caste and Mate Selection in Modern India,” *American Economic Journal: Microeconomics*, may 2013, 5 (2), 33–72.
- Barron, Kai**, “Belief Updating: Does the Good-News, Bad-News Asymmetry Extend to Purely Financial Domains?,” *Experimental Economics*, March 2021, 24 (1), 31–58.
- Bauer, Kevin, Yan Chen, Florian Hett, and Michael Kosfeld**, “Group Identity and Belief Formation: A Microfoundation of Political Polarization,” *The Economic Journal*, August 2026. Accepted manuscript published online 17 August 2026.
- Becker, Gary S.**, *The Economics of Discrimination*, Chicago: University of Chicago Press, 1957.
- Benjamin, Daniel J.**, “Errors in probabilistic reasoning and judgment biases,” in “Handbook of Behavioral Economics - Foundations and Applications 2,” Elsevier, 2019, pp. 69–186.
- BenYishay, Ariel and A. Mushfiq Mobarak**, “Social Learning and Incentives for Experimentation and Communication,” *The Review of Economic Studies*, May 2019, 86 (3), 976–1009.
- Berger, Sebastian, Christoph Feldhaus, and Axel Ockenfels**, “A shared identity promotes herding in an information cascade game,” *Journal of the Economic Science Association*, may 2018, 4 (1), 63–72.

- Bertrand, Marianne and Esther Duflo**, “Field Experiments on Discrimination,” in Abhijit V. Banerjee and Esther Duflo, eds., *Handbook of Economic Field Experiments*, Vol. 1, Elsevier, 2017, chapter 8, pp. 309–393.
- Bikhchandani, Sushil, David Hirshleifer, Omer Tamuz, and Ivo Welch**, “Information Cascades and Social Learning,” *Journal of Economic Literature*, September 2024, 62 (3), 1040–1093.
- Bohren, J. Aislinn, Kareem Haggag, Alex Imas, and Devin G. Pope**, “Inaccurate Statistical Discrimination: An Identification Problem,” *The Review of Economics and Statistics*, May 2025, 108 (3), 605–620.
- Boyd, Robert, Peter J. Richerson, and Joseph Henrich**, “The cultural niche: Why social learning is essential for human adaptation,” *Proceedings of the National Academy of Sciences*, jun 2011, 108 (supplement_2), 10918–10925.
- Cassan, Guilhem, Daniel Keniston, and Tatjana Kleineberg**, “A Division of Laborers: Identity and Efficiency in India,” NBER Working Paper 28462, National Bureau of Economic Research February 2021. Revised February 2022.
- Çelen, Boğaçhan, Shachar Kariv, and Andrew Schotter**, “An Experimental Test of Advice and Social Learning,” *Management Science*, October 2010, 56 (10), 1687–1701.
- Census of India**, “Census of India 2011: Provisional Population Totals,” Technical Report, Office of the Registrar General and Census Commissioner, India 2011. Reference ID ORGI_PPT_2011_31.
- Chakravarty, Surajeet, Miguel A. Fonseca, Sudeep Ghosh, and Sugata Marjit**, “Religious fragmentation, social identity and cooperation: Evidence from an artefactual field experiment in India,” *European Economic Review*, November 2016, 90, 265–279.
- Charness, Gary and Yan Chen**, “Social Identity, Group Behavior, and Teams,” *Annual Review of Economics*, aug 2020, 12 (1), 691–713.
- Chen, Daniel L., Martin Schonger, and Chris Wickens**, “oTree—An open-source platform for laboratory, online, and field experiments,” *Journal of Behavioral and Experimental Finance*, mar 2016, 9, 88–97.
- Chen, Yan and Sherry Xin Li**, “Group Identity and Social Preferences,” *American Economic Review*, March 2009, 99 (1), 431–457.
- Conlon, John J., Malavika Mani, Gautam Rao, Matthew W. Ridley, and Frank Schilbach**, “Learning in the Household,” NBER Working Paper 28844, National Bureau of Economic Research May 2021.
- , —, —, —, —, and —, “Not Learning from Others,” NBER Working Paper 30378, National Bureau of Economic Research August 2022. Revised January 2026.

- Danz, David, Lise Vesterlund, and Alistair J. Wilson**, “Belief Elicitation and Behavioral Incentive Compatibility,” *American Economic Review*, September 2022, 112 (9), 2851–2883.
- Dekel, Inbal and Moses Shayo**, “Follow, Ignore, or Counteract: On Social Identity, Social Learning, and Polarization,” November 2025. SSRN working paper, posted 13 June 2025 and last revised 20 November 2025.
- Dhami, Sanjit, Mengxing Wei, and Pavan Mamidi**, “Religious identity, trust, reciprocity, and prosociality: Theory and evidence,” *Journal of Development Economics*, January 2024, 166, 103192.
- Dietvorst, Berkeley J., Joseph P. Simmons, and Cade Massey**, “Algorithm aversion: People erroneously avoid algorithms after seeing them err,” *Journal of Experimental Psychology: General*, 2015, 144 (1), 114–126.
- , —, and —, “Overcoming Algorithm Aversion: People Will Use Imperfect Algorithms If They Can (Even Slightly) Modify Them,” *Management Science*, mar 2018, 64 (3), 1155–1170.
- Dimant, Eugen, Fabio Galeotti, and Marie Claire Villeval**, “Motivated Information Acquisition and Social Norm Formation,” *European Economic Review*, August 2024, 167, 104778.
- Dolan, P., M. Hallsworth, D. Halpern, D. King, R. Metcalfe, and I. Vlaev**, “Influencing behaviour: The mindspace way,” *Journal of Economic Psychology*, February 2012, 33 (1), 264–277.
- Enke, Benjamin and Thomas Graeber**, “Cognitive Uncertainty,” *The Quarterly Journal of Economics*, may 2023, 138 (4), 2021–2067.
- Falk, Armin and Florian Zimmermann**, “Information Processing and Commitment,” *The Economic Journal*, August 2018, 128 (613), 1983–2002.
- Fershtman, C. and U. Gneezy**, “Discrimination in a Segmented Society: An Experimental Approach,” *The Quarterly Journal of Economics*, February 2001, 116 (1), 351–377.
- Gangadharan, Lata, Pushkar Maitra, Joseph Vecchi, Prakashan Chellattan Veetil, and Marie Claire Villeval**, “Do Advisors’ Status and Identity Shape Adherence to Advice?,” IZA Discussion Paper 18280, IZA Institute of Labor Economics November 2025.
- García-Hombrados, Jorge, Marcel Jansen, Ángel Martínez, Berkay Özcan, Pedro Rey-Biel, and Antonio Roldán-Monés**, “Ideological Alignment and Evidence-Based Policy Adoption,” *American Economic Review*, July 2026, 116 (7), 2574–2603.
- Ghosh, Arkadev**, “Religious Divisions and Production Technology: Experimental Evidence from India,” *Journal of Political Economy*, October 2025, 133 (10), 3249–3304.
- Goeree, Jacob K. and Leeat Yariv**, “Conformity in the lab,” *Journal of the Economic Science Association*, feb 2015, 1 (1), 15–28.

- Guryan, Jonathan and Kerwin Kofi Charles**, “Taste-Based or Statistical Discrimination: The Economics of Discrimination Returns to its Roots,” *The Economic Journal*, nov 2013, 123 (572), F417–F432.
- Hartzmark, Samuel M., Samuel D. Hirshman, and Alex Imas**, “Ownership, Learning, and Beliefs,” *The Quarterly Journal of Economics*, August 2021, 136 (3), 1665–1717.
- Henrich, Joseph**, *The Secret of Our Success*, Princeton University Press, dec 2016.
- Hoff, Karla, Mayuresh Kshetramade, and Ernst Fehr**, “Caste and Punishment: The Legacy of Caste Culture in Norm Enforcement,” *The Economic Journal*, November 2011, 121 (556), F449–F475.
- Hou, Yoyo Tsung-Yu and Malte F. Jung**, “Who is the Expert? Reconciling Algorithm Aversion and Algorithm Appreciation in AI-Supported Decision Making,” *Proceedings of the ACM on Human-Computer Interaction*, oct 2021, 5 (CSCW2), 1–25.
- Itoh, Hideshi and Kimiyuki Morita**, “Information Acquisition, Decision Making, and Implementation in Organizations,” *Management Science*, January 2023, 69 (1), 446–463.
- Iyer, Sriya**, “The New Economics of Religion,” *Journal of Economic Literature*, June 2016, 54 (2), 395–441.
- **and Anand Shrivastava**, “Religious riots and electoral politics in India,” *Journal of Development Economics*, March 2018, 131, 104–122.
- Jaffrelot, Christophe**, *Modi’s India: Hindu Nationalism and the Rise of Ethnic Democracy*, Princeton, NJ: Princeton University Press, 2021.
- Lakens, Daniël, Anne M. Scheel, and Peder M. Isager**, “Equivalence Testing for Psychological Research: A Tutorial,” *Advances in Methods and Practices in Psychological Science*, June 2018, 1 (2), 259–269.
- Langer, Markus, Tim Hunsicker, Tina Feldkamp, Cornelius J. König, and Nina Grgić-Hlača**, ““Look! It’s a Computer Program! It’s an Algorithm! It’s AI!”: Does Terminology Affect Human Perceptions and Evaluations of Algorithmic Decision-Making Systems?,” in “CHI Conference on Human Factors in Computing Systems” CHI ’22 ACM April 2022, pp. 1–28.
- Levy, Gilat and Ronny Razin**, “Echo Chambers and Their Effects on Economic and Political Outcomes,” *Annual Review of Economics*, aug 2019, 11 (1), 303–328.
- Logg, Jennifer M., Julia A. Minson, and Don A. Moore**, “Algorithm Appreciation: People Prefer Algorithmic to Human Judgment,” *Organizational Behavior and Human Decision Processes*, March 2019, 151, 90–103.
- Lowe, Matt**, “Types of Contact: A Field Experiment on Collaborative and Adversarial Caste Integration,” *American Economic Review*, June 2021, 111 (6), 1807–1844.

- Maclean, Johanna Catherine, John Buckell, and Joachim Marti**, “Information Source and Cigarettes: Experimental Evidence on the Messenger Effect,” NBER Working Paper 25632, National Bureau of Economic Research March 2019.
- Mobius, Markus and Tanya Rosenblat**, “Social Learning in Economics,” *Annual Review of Economics*, aug 2014, 6 (1), 827–847.
- Mosse, David**, “Caste and Development: Contemporary Perspectives on a Structure of Discrimination and Advantage,” *World Development*, October 2018, 110, 422–436.
- , “The Modernity of Caste and the Market Economy,” *Modern Asian Studies*, July 2020, 54 (4), 1225–1271.
- Munshi, Kaivan**, “Caste and the Indian Economy,” *Journal of Economic Literature*, dec 2019, 57 (4), 781–834.
- Oh, Suanna**, “Does Identity Affect Labor Supply?,” *American Economic Review*, aug 2023, 113 (8), 2055–2083.
- Parys, Jessica Van and Elliott Ash**, “Sequential decision-making with group identity,” *Journal of Economic Psychology*, dec 2018, 69, 1–18.
- Pew Research Center**, “Religion in India: Tolerance and Segregation,” Technical Report, Pew Research Center June 2021. Report released 29 June 2021.
- Phelps, Edmund S.**, “The Statistical Theory of Racism and Sexism,” *American Economic Review*, September 1972, 62 (4), 659–661.
- Robbett, Andrea, Lily Colón, and Peter Hans Matthews**, “Partisan political beliefs and social learning,” *Journal of Public Economics*, apr 2023, 220, 104834.
- Sankaran, Sindhuja, Maciek Sekerdej, and Ulrich von Hecker**, “The Role of Indian Caste Identity and Caste Inconsistent Norms on Status Representation,” *Frontiers in Psychology*, mar 2017, 8.
- Shayo, Moses**, “A Model of Social Identity with an Application to Political Economy: Nation, Class, and Redistribution,” *American Political Science Review*, May 2009, 103 (2), 147–174.
- , “Social Identity and Economic Policy,” *Annual Review of Economics*, aug 2020, 12 (1), 355–389.
- Siddique, Zahra**, “Evidence on Caste Based Discrimination,” *Labour Economics*, dec 2011, 18, S146–S159.
- Tajfel, Henri and John C. Turner**, “An integrative theory of intergroup conflict,” 1979, pp. 33–47.
- , **M. G. Billig, R. P. Bundy, and Claude Flament**, “Social categorization and intergroup behaviour,” *European Journal of Social Psychology*, 1971, 1 (2), 149–178.
- Thorat, Sukhadeo and Paul Attewell**, “The Legacy of Social Exclusion: A Correspondence Study of Job Discrimination in India,” *Economic and Political Weekly*, 2007, 42 (41), 4141–4145.

Wabitsch, Alena, “The Messenger Matters,” October 2024. Working paper, posted May 2025 and last revised 20 November 2025.

Weizsäcker, Georg, “Do We Follow Others when We Should? A Simple Test of Rational Expectations,” *American Economic Review*, dec 2010, *100* (5), 2340–2360.

Zhang, Yunhao and David G. Rand, “Sincere or motivated? Partisan bias in advice-taking,” *Judgment and Decision Making*, 2023, *18*.

Zou, Wenbo and Xue Xu, “Ingroup Bias in a Social Learning Experiment,” *Experimental Economics*, March 2023, *26* (1), 27–54.